Scheduled Knowledge Distillation and Operational Threshold Optimization for Lightweight Medical Image Classification

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Abstract

In mobile and edge deployment of medical-image classifiers, real-world utility hinges on behavior at fixed per-class decision thresholds and on calibration—rather than AUROC alone. We therefore evaluate scheduled knowledge distillation (CE→KD) on ChestMNIST. A MobileNetV3-Small student (ImageNet-pretrained) is trained with a ResNet-18 teacher under a unified protocol; the KD mixing coefficient and temperature increase linearly (α: 0.1→0.7, τ: 2.0→5.0). Validation determines per-class F1-optimal thresholds, which are frozen for test. Reliability is assessed with 10-bin Expected Calibration Error (ECE) and reliability diagrams. Across three seeds, the supervised student attains macro-F1 0.2285±0.0071, ECE 0.0066±0.0027, AUROC 0.7796; the distilled student reaches macro-F1 0.2235±0.0044, ECE 0.3323±0.0039, AUROC 0.7669, exhibiting pronounced over-confidence. Analysis suggests a combination of teacher miscalibration transfer (amplified by increasing α and τ) and KD-only inverse class weighting that may dilute rare-positive signals. We release a fully reproducible pipeline (scripts, checkpoints, threshold files) via an anonymous artifact (Anonymous 2025).

1 Introduction

Deploying medical image classifiers to mobile and edge environments hinges on how predicted probabilities are converted into binary clinical decisions. Two axes dominate quality under this constraint: the F1 balance at fixed, per-class thresholds that cannot be tuned post-deployment, and calibration—how well predicted probabilities reflect empirical accuracy (e.g., Expected Calibration Error, ECE). Yet much of the knowledge-distillation (KD) literature emphasizes single-label settings, a single global threshold, and relative metrics such as AUROC/AUPRC, leaving a gap in multi-label medical tasks where F1 and ECE must be compared under frozen operational thresholds.

We address this gap on ChestMNIST by fixing the student as MobileNetV3-Small and applying scheduled KD from a ResNet-18 teacher, with αincreased linearly from 0.1 to 0.7 and temperature $\tau$from 2.0 to 5.0. Our evaluation protocol mirrors deployment: (i) select per-class, F1-optimal thresholds on validation; (ii) carry them unchanged to the test set; and (iii) report ECE (10 bins) and reliability diagrams alongside AUROC/AUPRC. Under this setting, the supervised student outperforms the distilled student in both macro-F1 (0.2285±0.0071 vs. 0.2235±0.0044) and calibration (ECE 0.0066±0.0027 vs. 0.3323±0.0039), indicating systematic over-confidence with KD despite similar F1.

Our contributions are threefold. First, we present a deployment-centric benchmark that fixes per-class thresholds and jointly reports macro-F1 and ECE. Second, we provide evidence that a simple CE→KD schedule with inverse class weighting applied only to the KD term can degrade calibration in multi-label medical classification. Third, we release a fully reproducible pipeline (scripts, logs, checkpoints, thresholds, and figures) via an anonymous artifact. This analysis complements AAAI work on distillation schedules and calibration methods—e.g., teacher-assistant distillation for bridging capacity gaps (Mirzadeh et al., 2020) and temperature-scaling-based calibration (Guo et al., 2017)—by reframing evaluation around fixed-threshold operational metrics rather than purely rank-based criteria.

2 Related Work

2.1 Knowledge Distillation and Training Procedures

Knowledge distillation (KD) transfers the teacher's soft-target distribution to a compact student to improve generalization (Hinton et al., 2015). Subsequent work refined training procedures to stabilize optimization and enhance transfer. Teacher-Assistant KD inserts an intermediate-capacity assistant to bridge teacher–student gaps and yields consistent gains when capacity disparity is large (Mirzadeh et al., 2020). Selective or reweighted KD variants pass only the most informative teacher signals, improving student quality in settings such as non-autoregressive translation (e.g., Liu et al., 2023). In this study we adopt a simple, implementation-friendly linear schedule that increases both the mixing coefficient and temperature, and we evaluate its effect on deployment-oriented metrics under fixed thresholds.

2.2 Multi-Label Classification and Long-Tail Imbalance

Multi-label medical imaging exhibits rare positives and uneven prevalence across classes, making average AUC insufficient for operational assessment. Cost-sensitive and reweighting schemes—such as effective-number class-balanced loss (Cui et al., 2019)—and margin-modifying objectives like Focal Loss and Asymmetric Loss (ASL) emphasize rare positives and hard negatives (Lin et al. 2017; Ben-Baruch et al. 2021). In parallel, multi-label-aware KD frameworks consider label-wise correlations and non-exclusive probabilities. We follow a minimal design that applies inverse class weights to the KD term only and quantify its operational effect under fixed thresholds and ECE reporting.

2.3 Calibration and Evaluation under Operational Thresholds

Calibration quantifies how faithfully predicted probabilities match empirical accuracy and is critical for clinical decision support. Post-hoc temperature scaling remains a strong baseline (Guo et al., 2017), while recent AAAI work explores adaptive or sample-dependent temperatures and improved estimators of calibration. Reliability diagrams and ECE are standard, but threshold-sensitive metrics require careful handling and evaluation protocols that lock thresholds before testing. Following these recommendations, we select per-class thresholds on validation, freeze them for test, and jointly report macro-F1 and ECE. Positioning. Prior AAAI contributions refined KD schedules, assistant-based distillation, representation transfer, long-tail objectives, and calibration; few, however, fix a lightweight student, combine scheduled KD with class weighting, and evaluate both macro-F1 and ECE under frozen per-class thresholds on a multi-label medical task. We fill this gap with a simple, reproducible pipeline and report three-seed means and standard deviations to quantify deployment-relevant effects.

3 Method

3.1 Task, Models, and Objective

We study multi-label classification with 14 thoracic findings on the official ChestMNIST split (Yang et al. 2023). The student backbone is MobileNetV3 (Howard et al. 2019) and the teacher is ResNet-18 (He et al. 2016). Optimization uses AdamW (Loshchilov and Hutter 2019). The classifier head applies a per-class sigmoid, producing $\mathrm{p_S} = \sigma(\mathrm{z_S}) \in [0,1]^{\mathrm{C}}$ $(\mathrm{C} = 14)$.

The training objective combines supervised BCE with knowledge distillation (KD):

$$\mathcal{L} = (1-\alpha)\,\mathcal{L}_{\mathrm{BCE}} + \alpha\,\mathcal{L}_{\mathrm{KD}},$$

$$\mathcal{L}_{\mathrm{BCE}} = \frac{1}{C}\sum_{c=1}^{C} \mathrm{BCE}\,(p_{S,c}, y_c),$$

$$\mathcal{L}_{\mathrm{KD}} = \frac{1}{C}\sum_{c=1}^{C} \quad w_c\,\mathrm{KL}\,(\sigma(z_{T,c}/\tau)\ \parallel\ \sigma(z_{S,c}/\tau)).$$

Here w_c is applied only to the KD term; we intentionally do not use the τ^2 scaling correction.

3.2 Training Procedure and Schedules

All models are trained with AdamW (learning rate 3×10^{-4}, weight decay 10^{-4}), batch size 64, for 12 epochs (Loshchilov and Hutter 2019). Data augmentation is light—random horizontal flip plus small translation/crop. Early stopping uses patience 5 on a validation metric. We repeat all experiments with seeds $\{0,1,2\}$.

Scheduled KD (CE→KD) linearly increases both α and τ over epochs $e \in \{1,\dots,E\}$:

$$\alpha(e) = \alpha_{\min} + (\alpha_{\max} - \alpha_{\min})\frac{e-1}{E-1}, \tau(e) = \tau_{\min} + (\tau_{\max} - \tau_{\min})\frac{e-1}{E-1},$$

with $\alpha_{\min} = 0.1$, $\alpha_{\max} = 0.7$, $\tau_{\min} = 2.0$, $\tau_{\max} = 5.0$. For class imbalance, we use inverse class weighting only in KD:

$$w_c = \frac{1}{\pi_c + \varepsilon},$$

where π_c is the training prevalence of class c and ε a small constant. The supervised BCE is left unweighted.

3.3 Evaluation Protocol

(i) Validation-time threshold search.

$\theta_c^* = \arg\max\ \theta \in [0,1]\ F1_c(\theta)$,

using a grid $[0.00, 1.00]$ with step 0.01 (independent per class).

(ii) Fixed-threshold testing.

Freeze $\{\theta_c^*\}_{c=1}^{C}$ and apply them unchanged to the test set to compute macro-F1.

(iii) Calibration.

$$\mathrm{ECE} = \sum_{m=1}^{M} \frac{|\,B_m\,|}{N}\ |\,\mathrm{acc}(B_m) - \mathrm{conf}(B_m)\,|, M = 10,$$

plus reliability diagrams. All evaluations use the fixed-threshold regime. The pipeline saves thresholds (JSON), calibration plots, checkpoints, and logs. The full fixed-threshold protocol is summarized in Algorithm 1.

4 Experiments

Algorithm 1: Fixed-Threshold Evaluation

Input: training set D_{train}, validation set D_{val}, test set D_{test}; teacher T, student S; schedules $\alpha(e), \tau(e)$; class weights w_c (KD only).

Training: For epochs $e = 1,\dots,E$:
a. Update S to minimize $(1-\alpha(e))\,L_{\mathrm{BCE}} + \alpha(e)\,L_{\mathrm{KD}}(\tau(e), w)$, where
$L_{\mathrm{KD}} = \frac{1}{C}\sum_{c=1}^{C} \quad w_c\,\mathrm{KL}\,(\sigma(z_c^T/\tau(e))\ \parallel\ \sigma(z_c^S/\tau(e)))$.
b. Track validation macro-F1/AUPRC; save best checkpoint.

Validation-time threshold search: For each class $c = 1,\dots,C$:
a. Sweep $\theta \in \{0.00, 0.01, \dots, 1.00\}$;
b. Compute $F1_c(\theta)$ on D_{val};
c. Set $\theta_c^* = \arg\max_\theta F1_c(\theta)$.
Save $\{\theta_c^*\}$ as *thresholds_val_f1.json*.

Testing (frozen thresholds):
Load checkpoint and $\{\theta_c^*\}$. On D_{test}, compute macro-F1 using θ_c^* per class.
Compute ECE with $M = 10$ equal-width bins:
$\mathrm{ECE} = \sum_{m=1}^{M} \frac{|B_m|}{N}\ |\,\mathrm{acc}(B_m) - \mathrm{conf}(B_m)\,|$.
Save logs, reliability plots, and metrics.

4.1 Setup

All experiments use an identical configuration. Inputs are resized to 128×128, batch size 64, 12 training epochs, and AdamW with learning rate $3 \times 10^{-4}$and weight decay 10^{-4}. Augmentation is light (random horizontal flip and small translation/crop). Each experiment is repeated with seeds $\{0,1,2\}$. The student is MobileNetV3-Small initialized from ImageNet; the teacher is ResNet-18. Per-label probabilities are produced with class-wise sigmoids.

The total loss is

$L = (1-\alpha)\, L_{\text{BCE}} + \alpha\, L_{\text{KD}}, L_{\text{KD}} = \text{KL}(\sigma(z_T/\tau) \parallel \sigma(z_S/\tau))$.

Distillation uses a linear CE→KD schedule with $\alpha: 0.1 \rightarrow 0.7$and $\tau: 2.0 \rightarrow 5.0$. Inverse class weighting is applied only to the KD term; the supervised BCE is left unweighted.

The evaluation mirrors deployment: (i) on validation, grid-search per-class thresholds $\{\theta_c^*\}_{c=1}^{14}$ that maximize class-wise F1 (0.00–1.00, step 0.01); (ii) freeze those thresholds for the test set; and (iii) quantify calibration with 10-bin ECE and reliability diagrams. For each run, the pipeline saves best.pt, run.log, thresholds_val_f1.json, and calibration plots. In a teacher-only calibration check, several classes yielded degenerate thresholds converging to 0.0; we therefore exclude teacher calibration interpretation and use the teacher only for distillation.

4.2 Main Results

Three-seed means ± standard deviations

Student (supervised) AUROC 0.7796 ± 0.0031, AUPRC 0.1711 ± 0.0038, macro-F1 0.2285 ± 0.0071, ECE 0.0066 ± 0.0027

Distill (CE→KD, inverse@KD) AUROC 0.7669 ± 0.0139, AUPRC 0.1640 ± 0.0011, macro-F1 0.2235 ± 0.0044, ECE 0.3323 ± 0.0039

Teacher (reference) AUROC ≈ 0.766–0.767, AUPRC ≈ 0.157–0.159, macro-F1 ≈ 0.211–0.213

Method	AUROC	AUPRC	Macro-F1	ECE
Student	0.7796 ± 0.0031	0.1711 ± 0.0038	0.2285 ± 0.0071	0.0066 ± 0.0027
Distill	0.7669 ± 0.0139	0.1640 ± 0.0011	0.2235 ± 0.0044	0.3323 ± 0.0039

Table 1. Main results (mean ± s.d.)

Method	s0	s1	s2
Student	0.2230 / 0.0097	0.2365 / 0.0048	0.2260 / 0.0053
Distill	0.2253 / 0.3298	0.2184 / 0.3368	0.2267 / 0.3303

Table 2. Per-seed macro-F1 / ECE

Calibration diverges markedly between models **(**Fig. 1**):** the distilled model is over-confident across bins (ECE ≈ 0.33), whereas the supervised student aligns closely with the diagonal (ECE ≈ 0.007).

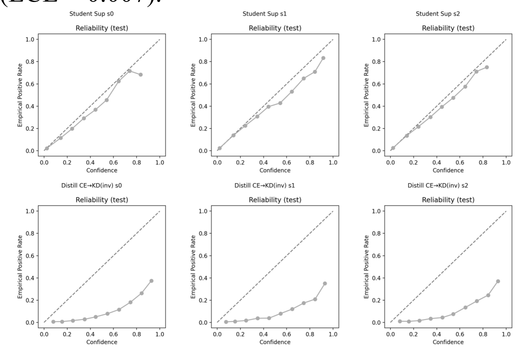

Figure 1. Reliability diagrams under frozen per-class thresholds (M = 10 bins).

Student (supervised) shows near-diagonal alignment (ECE ≈ 0.007). Right: Distill (CE→KD, inverse@KD) exhibits systematic over-confidence across bins (ECE ≈ 0.33). Shaded bars indicate per-bin sample density.

4.3 Analysis

The supervised student shows a clear operational advantage: with validation-optimal thresholds frozen at test time, it achieves higher macro-F1 and markedly lower ECE (≈ 0.007). AUROC/AUPRC are also consistently stronger, suggesting that the pretrained student representation is robust and that the present KD signal does not yield fixed-threshold gains. The distilled model exhibits systematic over-confidence (ECE ≈ 0.33), with reliability curves that overstate accuracy. Likely causes include transfer of teacher miscalibration amplified by the increasing $\alpha$and τschedule, and attenuation of rare-positive signals because inverse weighting is applied only to the KD term under a high temperature. Sensitivity on extremely low-prevalence classes is weaker under distillation, indicating that restricting reweighting to the KD term alone appears insufficient to counterbalance hard-label supervision, especially for extremely low-prevalence classes. This motivates weighted supervision, alternative losses (ASL/Focal), and feature-level distillation (Zagoruyko and Komodakis 2017). Overall, in the evaluated setting (ResNet-18 teacher, linear CE→KD schedule, inverse@KD), distillation does not improve fixed-threshold performance and degrades calibration.

5 Discussion / Limitations / Ethics

5.1 Discussion

From an operational perspective, our setting—ResNet-18 teacher, a linear CE→KD schedule, and inverse class weighting applied only to the KD term—yields a consistent pattern. The supervised student achieves higher test macro-F1 (0.2285 ± 0.0071)and markedly lower ECE (0.0066 ± 0.0027), whereas the distilled student attains a similar macro-F1 but substantially worse calibration (ECE =

0.3323 ± 0.0039), indicating systematic over-confidence. Under fixed thresholds, such miscalibration can inflate the cost of both false alarms and missed detections.

We hypothesize three non-exclusive mechanisms. First, miscalibration transfer from the teacher as $\alpha$and τincrease, which injects teacher probability patterns more strongly in later training and encourages over-confident logits. Second, rare-class signal dilution because inverse weighting is applied only to the KD term and higher temperatures ($\tau: 2 \rightarrow 5$)flatten soft targets, weakening decision boundaries for infrequent positives. Third, limited headroom with a strong ImageNet-pretrained student, leaving little opportunity for KD to improve thresholded performance without harming confidence alignment.

These findings motivate deployment-centric recommendations. Model selection should fix per-class thresholds on validation and evaluate on test without re-tuning; rank-based metrics alone may mask calibration collapse. Thresholds are sensitive to distribution shifts across sites, devices, and seasons, so periodic re-tuning and drift detection are necessary. If KD is used, pair it with stronger calibration control and loss design, such as teacher temperature scaling before distillation, non-linear (α, τ) schedules, joint reweighting of supervision and KD, feature-level distillation, and post-hoc temperature scaling followed by renewed threshold search. We further recommend a reporting standard that includes fixed validation-optimal thresholds, ECE with 10 bins, and reliability diagrams. During training, consider temperature scaling for both teacher and student, explore non-linear schedules, and address class imbalance by weighting the supervised BCE or replacing it with ASL/Focal. For rare classes, per-class αor targeted rare-class reweighting can stabilize decision boundaries.

5.2 Limitations

ChestMNIST is a low-resolution derived benchmark that does not reflect clinical resolution or multi-institutional diversity. We used a single source and the official split without external validation; label noise may be present. Methodologically, we used one teacher architecture and a linear CE→KD schedule; alternatives were not systematically explored. Inverse weighting was applied only to KD; alternatives such as ASL/Focal or simultaneous weighting of the supervised loss were not tested. Feature-level distillation was disabled. For evaluation, we reported ECE (10 bins) but not Brier, NLL, or MCE; ECE can be sensitive to binning. Threshold search used a 0.01 grid; finer granularity may yield small differences. Statistical testing was limited, and a few teacher-only classes produced degenerate thresholds (0.0), which were excluded. Efficiency metrics (latency, memory, energy) were not reported.

5.3 Ethics

This work is for educational and benchmarking purposes only and must not be used directly for clinical decision-making. Prior to deployment, IRB oversight, regulatory authorization, and multi-site prospective validation are required. Model outputs should be used strictly as decision support with mandatory second reads for high-risk cases. Validation-derived thresholds should be documented with audit logs and versioning for any changes. Continuous monitoring for performance and distribution drift, along with safe rollback paths for thresholds and models, is essential. Task-specific cost models for false negatives and false positives should be articulated and reflected in triage policies. Fairness and bias should be assessed via stratified performance and ECE across demographic groups, institutions, and devices, with disparities addressed by differentiated thresholds or recalibration. Privacy and security require de-identification, access control, least-privilege, and encryption. Public releases should state license terms and usage restrictions. Core artifacts and execution scripts should be released together, and the model card should state intended use, prohibited use, and known limitations.

6 Conclusion

We presented a deployment-oriented evaluation of scheduled CE→KD on multi-label ChestMNIST with a fixed lightweight student and a ResNet-18 teacher. By selecting per-class F1-optimal thresholds on validation and freezing them for test, and by jointly reporting macro-F1 with calibration metrics, we modeled mobile and edge constraints. The supervised student consistently outperformed the distilled student in both macro-F1 and calibration; the distilled model showed pronounced over-confidence despite similar F1. Evidence suggests teacher miscalibration transfer, temperature-induced flattening that dilutes rare-class signals, and limited headroom given a strong pretrained student. Consequently, basic scheduled KD with KD-only reweighting does not guarantee operational gains and may degrade calibration. Future work should combine distillation with temperature scaling for teacher and student, non-linear schedules, reweighted supervision or alternative losses, and feature-level guidance. For deployment-ready models, fixed per-class thresholds, explicit ECE reporting, online monitoring, and rollback procedures are essential.

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